Indian Institute of Technology Dharwad

CS201L: Artificial Intelligence Laboratory

Matplotlib Library : Functions

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Matplotlib-Visualization with Python

Matplotlib is a foundational Python library for creating static, animated, and interactive data visualizations like line plots, scatter plots, histograms, and charts, offering extensive customization and output to various formats, essential for data science and analysis. It works seamlessly with NumPy and Pandas, often using matplotlib.pyplot (as plt) interface for easy plotting.

Table 1: Commonly Used Matplotlib Functions, Descriptions, and Parameters

Function	**Description**	**Parameters**
plt.figure()	Creates a new figure window.	`plt.figure(num=None, figsize=None, dpi=None, *, facecolor=None, edgecolor=None, frameon=True, clear=False, **kwargs)`
plt.plot()	Plots lines and/or markers to represent data points.	`plt.plot(*args, scalex=True, scaley=True, data=None, **kwargs)`
plt.scatter()	Creates a scatter plot.	`plt.scatter(x, y, s=None, c=None, *, marker=None, cmap=None, norm=None, vmin=None, vmax=None, alpha=None, linewidths=None, edgecolors=None, plotnonfinite=False, data=None, **kwargs)`

Function	Description	Parameters
plt.bar()	Draws a vertical bar chart.	`plt.bar(x, height, width=0.8, bottom=None, *, align='center', data=None, **kwargs)`
plt.barh()	Draws a horizontal bar chart.	`plt.barh(y, width, height=0.8, left=None, *, align='center', **kwargs)`
plt.hist()	Plots a histogram to visualize data distribution.	`plt.hist(x, bins=None, *, range=None, density=False, weights=None, cumulative=False, bottom=None, histtype='bar', align='mid', orientation='vertical', rwidth=None, log=False, color=None, label=None, stacked=False, data=None, **kwargs)`
plt.pie()	Displays a pie chart.	`plt.pie(x, *, explode=None, labels=None, colors=None, autopct=None, pctdistance=0.6, shadow=False, labeldistance=1.1, startangle=0, radius=1, counterclock=True, frame=False, rotatelabels=False, normalize=True, hatch=None, data=None)`

Function	**Description**	**Parameters**
plt.xlabel()	Sets the label for the x-axis.	`plt.xlabel(xlabel, fontdict=None, labelpad=None, *, loc=None, **kwargs)`
plt.ylabel()	Sets the label for the y-axis.	`plt.ylabel(ylabel, fontdict=None, labelpad=None, *, loc=None, **kwargs)`
plt.title()	Sets the title of the plot.	`plt.title(label, fontdict=None, loc=None, pad=None, *, y=None, **kwargs)`
plt.legend()	Displays the legend on the plot.	`plt.legend(*args, **kwargs)`
plt.grid()	Adds grid lines to the plot.	`plt.grid(visible= None, which='major', axis='both', **kwargs)`
plt.xlim()	Sets the limits of the x-axis.	`plt.xlim(*args, **kwargs)`
plt.ylim()	Sets the limits of the y-axis.	`plt.ylim(*args, **kwargs)`
plt.subplot()	Creates subplots in a figure.	`plt.subplots( nrows=1, ncols=1, *, sharex=False, sharey=False, squeeze=True)`
plt.tight_layout()	Automatically adjusts subplot parameters to fit within the figure.	`plt.tight_layout(*, pad=1.08, h_pad=None, w_pad=None, rect=None)`
plt.savefig()	Saves the current figure to a file.	`plt.savefig(*args, **kwargs)`
plt.show()	Displays the figure window.	`plt.show(*, block=None)`
plt.style.use()	Applies a predefined plotting style.	`plt.style.use( 'ggplot')`

Function	Description	Parameters
plt.fill_between()	Fills the area between two curves.	`plt.fill_between(x, y1, y2=0, where=None, interpolate=False, step=None, *, data=None, **kwargs)`
plt.errorbar()	Plots error bars on data points.	`plt.errorbar(x, y, yerr=None, xerr=None, fmt='', *, ecolor=None, elinewidth=None, capsize=None, barsabove=False, lolims=False, uplims=False, xlolims=False, xuplims=False, errorevery=1, capthick=None, data=None, **kwargs)`
plt.annotate()	Adds text annotations to specific points.	`plt.annotate(text, xy, xytext=None, xycoords='data', textcoords=None, arrowprops=None, annotation_clip=None, **kwargs)`
plt.clf()	Clears the current figure.	`plt.clf()`
plt.close()	Closes the current figure window.	`plt.close(fig=None)`

For details on operations in Matplotlib, see the official documentation: `https://matplotlib.org/stable/users/explain/quick_start.html#quick-start`